MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS)
THIRD SEMESTER EXAMINATION
(2024 ADMISSION ONWARDS)
COURSE CODE: MG3DSEFTQ200
COURSE TITLE: ENZYMES IN FOOD INDUSTRY

Duration:2hrs Maximum Marks: 70

Part A

Answer in one word
Each question carries **1** Mark

1. What is the function of enzymes. [R] [1]
2. Define GRAS in context with Enzyme application? [U] [2]
3. State the any two industrial sources of Lipases.[An] [3]
4. Name any two Proteolytic enzymes used in Dairy industry?[U][4]
5. How do Amylases aid in baking? [R] [2]
6. What is transglutaminase and how is it used in food processing?[A] [3]
7. What is downstream processing in enzyme production? [An][3]
8. Mention two advantages of enzymes in food industry over chemical catalysis.[E] [5]
9. What is the function of catalase in food packaging. [R] [1]
10. Define immobilized enzymes and give one food related example? [U] [5]

(10x1=10 marks)

Part B

Short Essay Type Questions
Answer any **6**questions
Each question carries **5** Marks

11. Describe the role of enzymes in brewing industry. [R] [2]
12. Explain the use of lipase in cheese industry [U] [4]
13. Discuss the importance of enzyme stability in foodprocessing [U] [5]
14. Outline the mechanism of action of cellulases in juice extraction. [A]3]
15. Briefly describe the steps in downstream processing of enzymes. [U][5]
16. Explain how enzymes are used in starch modification. [An][4]
17. Write detailed notes on:
    (a) Enzymes in antimicrobial coatings
    (b) Enzymes in flavor or nutrient-enhancing edible films
    (c) Regulatory assessment of enzymes in foods. [An] [5]
18. What are the ecological benefits of enzymatic food processing. [U][5]

(6x5=30 marks)

Part C

Long Essay Type Questions

Answer any **2** questions

Each question carries **15** Marks

19. Describe the use of proteases in meat tenderization, cheese making and fish processing.[U] [2]

20.Explain fermentative production of amylases used in the food industry and describe downstream processing steps. [U] [4]

21.Discuss the regulatory and safety aspects involvedin the use of enzymes in the food industry. [R] [3]

22. Explain the enzyme immobilization techniques and their advantages in continuous food processing. [A] [5]

(2x15=30 marks)